MERCK MANUAL
Professional Version

Parkinson Disease

(Parkinson's Disease)

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Parkinson disease is a slowly progressive, degenerative disorder characterized by resting tremor, stiffness (rigidity), slow and decreased movement (bradykinesia), and eventually gait and/or postural instability. Diagnosis is based on history and physical examination. Treatment aims to restore dopaminergic function in the brain with levodopa plus carbidopa and/or other medications (eg, dopamine agonists, monoamine oxidase type B [MAO-B] inhibitors, amantadine). For refractory symptoms in patients without dementia, stereotactic deep brain stimulation or lesional surgery and levodopa and an apomorphine pump may help.

(See also Overview of Movement and Cerebellar Disorders.)

The prevalence of Parkinson disease in North America is approximately 0.6% in adults age ≥ 45; prevalence increases with age (1).

The mean age at onset is approximately 60 years. Patients with tremor-dominant disease may develop motor symptoms earlier than those with akinetic-rigid disease (2).

Parkinson disease is usually idiopathic.

Juvenile parkinsonism, which is rare, begins during childhood or adolescence up to 20 years. Onset between ages 21 and 40 years is sometimes called young or early-onset Parkinson disease. Genetic causes are more likely in juvenile and early-onset Parkinson disease; these forms may differ from later-onset Parkinson disease because

- They progress more slowly.
- They are very sensitive to dopaminergic treatments.
- Most disability results from nonmotor symptoms such as depression, anxiety, and pain.

Secondary parkinsonism is brain dysfunction that is characterized by basal ganglia dopaminergic blockade and that is similar to Parkinson disease, but it is caused by something other than Parkinson disease (eg, medications, cerebrovascular disease, trauma, postencephalitic changes).

Atypical parkinsonism refers to a group of neurodegenerative disorders that have some features similar to those of Parkinson disease but have some different clinical features, a worse prognosis, a modest or no response to levodopa, and a different pathology (eg, neurodegenerative disorders such as multiple system atrophy, progressive supranuclear palsy, dementia with Lewy bodies, and corticobasal ganglionic degeneration).

General references

1. Willis AW, Roberts E , Beck JC, et al. Incidence of Parkinson disease in North America. *NPJ Parkinsons Dis.* 2022;8(1):170. doi: 10.1038/s41531-022-00410-y
2. Rajput AH, Voll A, Rajput ML, Robinson CA, Rajput A. Course in Parkinson disease subtypes: A 39-year clinicopathologic study. *Neurology.* 2009;73(3):206-212. doi: 10.1212/WNL.0b013e3181ae7af1

Pathophysiology of Parkinson Disease

Synuclein is primarily a neuronal cell protein that can aggregate into insoluble fibrils and form Lewy bodies.

The **pathologic hallmark of sporadic or idiopathic Parkinson disease** is

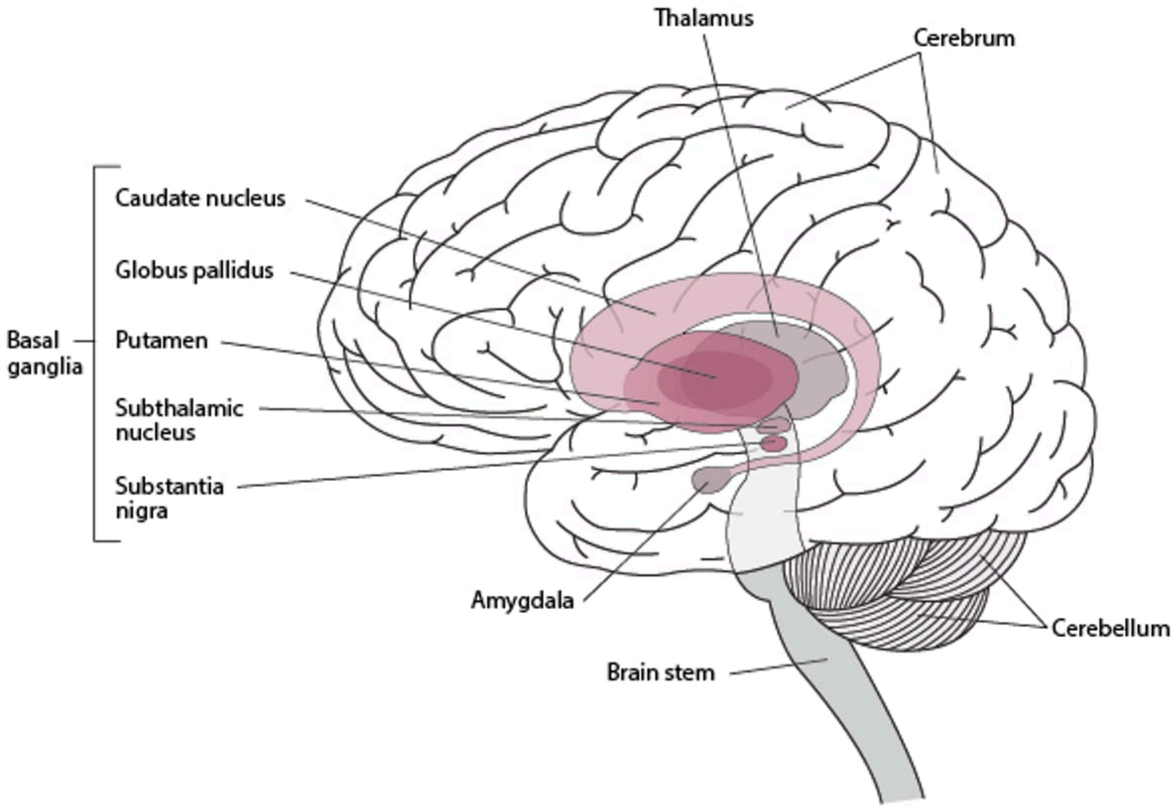
- Synuclein-filled Lewy bodies in the nigrostriatal system

However, synuclein can accumulate in many other parts of the nervous system, including the dorsal motor nucleus of the vagus nerve, basal nucleus of Meynert, hypothalamus, neocortex, olfactory bulb, sympathetic ganglia, and myenteric plexus of the gastrointestinal tract. Lewy bodies appear in a temporal sequence, and many experts believe that Parkinson disease is a relatively late development in a systemic synucleinopathy. Other synucleinopathies (synuclein deposition disorders) include dementia with Lewy bodies and multiple system atrophy. Parkinson disease may share features of other synucleinopathies, such as autonomic dysfunction and dementia.

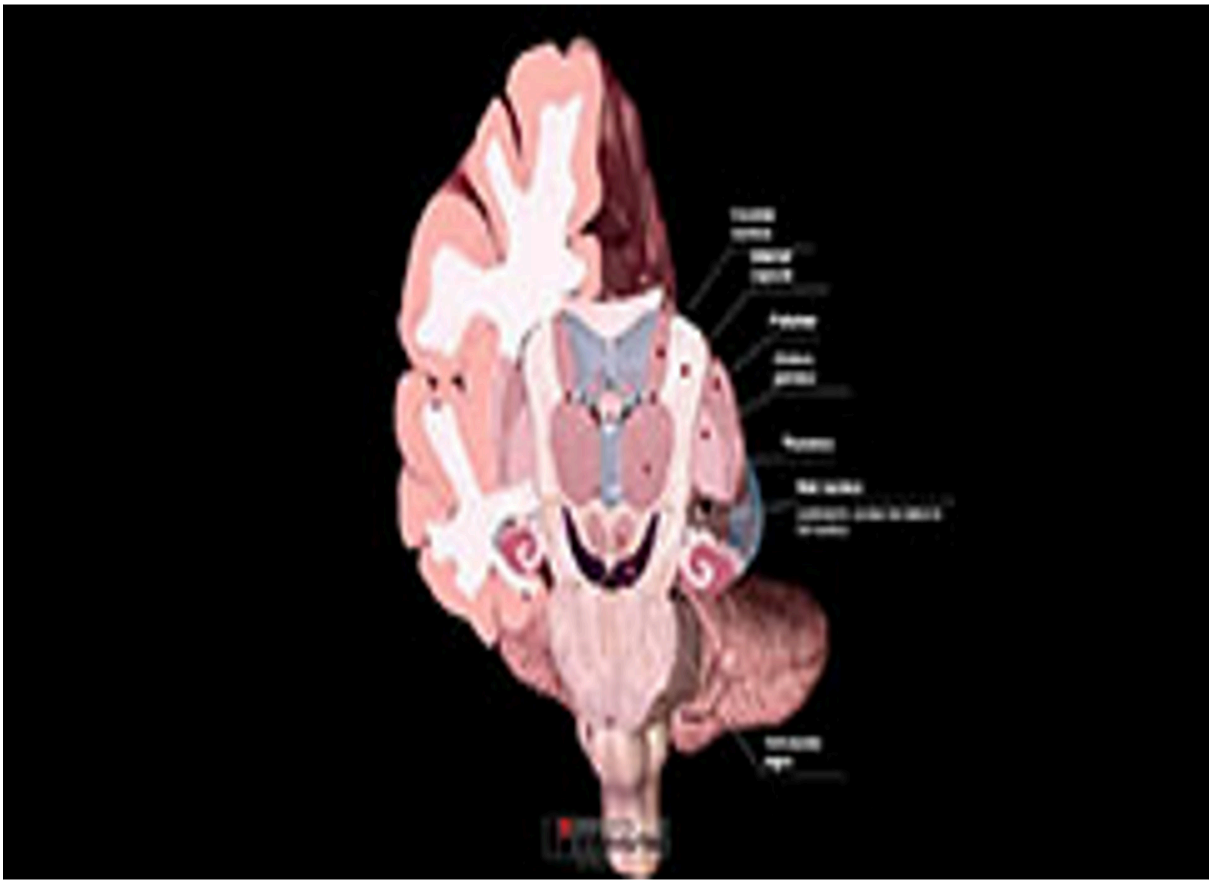
Rarely, Parkinson disease occurs without Lewy bodies (eg, in a form due to a mutation in the *PARK 2* gene).

In Parkinson disease, pigmented neurons of the substantia nigra, locus ceruleus, and other brain stem dopaminergic cell groups degenerate. Loss of substantia nigra neurons results in depletion of dopamine in the dorsal aspect of the putamen (part of the basal ganglia) and causes many of the motor manifestations of Parkinson disease (see figure Basal Ganglia).

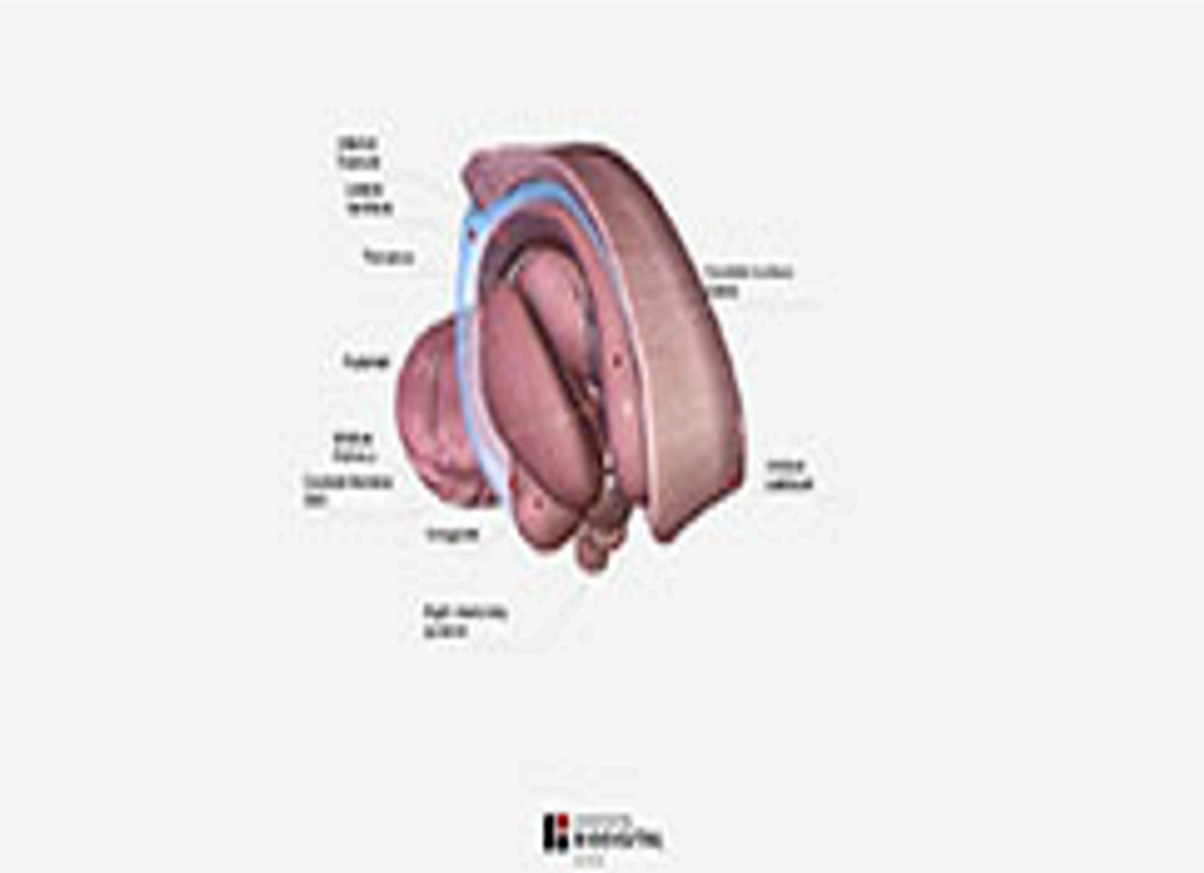
Basal Ganglia



Basal Ganglia: Coronal Section



Basal Ganglia: Regional Dissection



Etiology of Parkinson Disease

Although Parkinson disease was initially considered sporadic, accumulating evidence suggests that Parkinson disease has a substantial genetic component, with estimates of approximately 10 to 25% of disease risk due to genetic variation (1, 2). The frequency of genetic mutations varies by ethnicity.

Worldwide, mutations in *LRRK2* (encoding for leucine-rich repeat protein kinase 2) and *GBA* (encoding for glucocerebrosidase) are the 2 most common genetic determinants of Parkinson disease. Mutations, duplications, or triplications in the gene for alpha-synuclein are very rare, familial causes of Parkinson disease.

Environmental risk factors include exposure to carbon monoxide, pesticide use, and chronic consumption of well water. Caffeine intake, smoking, and physical activity predict lower risk (2).

Etiology references

1. Ye H, Robak LA, Yu M, Cykowski M, Shulman JM. Genetics and Pathogenesis of Parkinson's Syndrome. *Annu Rev Pathol.* 2023;18:95-121. doi:10.1146/annurev-pathmechdis-031521-034145
2. Ben-Shlomo Y, Darweesh S, Llibre-Guerra J, Marras C, San Luciano M, Tanner C. The epidemiology of Parkinson's disease. *Lancet.* 2024;403(10423):283-292. doi:10.1016/S0140-6736(23)01419-8

Symptoms and Signs of Parkinson Disease

In most patients, symptoms of Parkinson disease begin insidiously.

A **resting tremor** of 1 hand is often the first symptom. The tremor is characterized as follows:

- Slow and coarse
- Maximal at rest, lessening during movement, and absent during sleep
- Amplitude increased by emotional tension or fatigue
- Often involving the wrist and fingers, sometimes involving the thumb moving against the index finger (pill rolling), as when people roll a pill in their hand or handle a small object

Usually, the hands or feet are affected first, most often asymmetrically. The jaw and tongue may also be affected, but not the voice. However, speech may become hypophonic, with characteristic monotonous, sometimes stuttering dysarthria. Tremor may become less prominent as rigidity progresses. In predominantly rigid-akinetic forms of Parkinson disease, resting tremor is subtle or absent.

Rigidity develops independently of tremor in many patients. When a clinician moves a rigid joint, semirhythmic jerks occur because the intensity of the rigidity varies, causing a ratchet-like effect (cogwheel rigidity).

Slow movements (bradykinesia) are typical in Parkinson disease. Repetitive motor activity results in a progressive or sustained decrease in amplitude of movement (hypokinesia), and movement becomes hard to initiate (akinesia).

Rigidity and hypokinesia may contribute to muscle aches and sensations of fatigue. The face becomes masklike (hypomimic), with an open mouth and reduced blinking. Excessive drooling (sialorrhea) may contribute to disability.

Hypokinesia and impaired control of distal muscles cause micrographia (writing in very small letters) and make activities of daily living increasingly difficult.

Postural instability may develop later in Parkinson disease; if present at disease onset, alternative diagnoses should be suspected. Patients have difficulty starting to walk, turning, and stopping. They shuffle, taking short steps, holding their arms flexed to the waist, and swinging their arms little or not at all with each stride. Steps may inadvertently quicken, while stride length progressively shortens; this gait abnormality, called festination, is often a precursor to freezing of gait (when, without warning, walking and other voluntary movements may suddenly halt). A tendency to fall forward (propulsion) or backward (retropulsion) when the center of gravity is displaced results from loss of postural reflexes. Posture becomes stooped.

Dementia develops in approximately one-third of patients, usually late (1) in Parkinson disease. Early predictors of its development are visuospatial impairment (eg, getting lost while driving) and decreased verbal fluency.

Sleep disorders are common. Insomnia may result from nocturia or from the inability to turn in bed. Sleep deprivation may exacerbate depression and cognitive impairment and contribute to excessive daytime sleepiness. Rapid eye movement (REM) sleep behavior disorder may develop; in this disorder, verbalization and uncontrollable, possibly violent limb movements occur during REM sleep because the paralysis that normally occurs during REM sleep is absent. REM sleep behavior disorder is often accompanied by early neurodegenerative signs that occur primarily in patients with alpha-synucleinopathies, which can precede and/or increase the risk of developing Parkinson disease, multiple system atrophy, or dementia with Lewy bodies.

Neurologic symptoms unrelated to parkinsonism commonly develop because synucleinopathy occurs in other areas of the central, peripheral, and autonomic nervous systems. The following are examples:

- Sympathetic denervation of the heart, contributing to orthostatic hypotension in almost all patients
- Esophageal dysmotility, contributing to dysphagia and increased risk of aspiration
- Lower bowel dysmotility, contributing to constipation
- Urinary hesitancy and/or urgency, potentially leading to incontinence (common)
- Anosmia (common)

In some patients, some of these symptoms occur before the motor symptoms of Parkinson disease and frequently worsen over time.

Seborrheic dermatitis is also common (2).

Symptoms and signs references

1. Aarsland D, Creese B, Politis M, et al. Cognitive decline in Parkinson disease. *Nat Rev Neurol.* 2017;13(4):217-231. doi:10.1038/nrneurol.2017.27
2. Tomic S, Kuric I, Kuric TG, et al. Seborrheic Dermatitis Is Related to Motor Symptoms in Parkinson's Disease. *J Clin Neurol.* 2022;18(6):628-634. doi:10.3988/jcn.2022.18.6.628

Diagnosis of Parkinson Disease

- History and physical examination, mainly based on motor symptoms

Diagnosis of Parkinson disease is based on findings from history and physical examination. Parkinson disease is suspected in patients with characteristic unilateral resting tremor, decreased movement, or rigidity. During finger-to-nose coordination testing, the tremor disappears (or attenuates) in the limb being tested.

During the neurologic examination, patients cannot perform rapidly alternating or rapid successive movements well. Sensation and strength are usually normal. Reflexes are normal but may be difficult to elicit because of marked tremor or rigidity.

Slowed and decreased movement due to Parkinson disease must be differentiated from decreased movement and spasticity due to lesions of the corticospinal tracts. Unlike Parkinson disease, corticospinal tract lesions cause

- Paresis (weakness or paralysis), preferentially in distal antigravity muscles
- Hyperreflexia
- Extensor plantar responses (Babinski sign)
- Spasticity that increases muscle tone in proportion to the rate and degree of stretch placed on a muscle until resistance suddenly melts away (clasp-knife phenomenon)

The diagnosis of Parkinson disease is supported by the presence of other signs such as infrequent blinking, lack of facial expression, and gait abnormalities. Postural instability is also present, but if it occurs early in the disease, clinicians should consider other possible diagnoses.

In older adults, other possible causes of decreased spontaneous movements or a short-stepped gait must be excluded before Parkinson disease can be diagnosed; these include severe depression, hypothyroidism, or use of antipsychotics or certain antiemetics.

Parkinson disease should also be distinguished from secondary parkinsonism (most commonly due to medications or exogenous toxins). To help distinguish Parkinson disease from secondary or atypical parkinsonism, clinicians often test responsiveness to levodopa. A large, sustained response strongly supports Parkinson disease. A modest or no response to levodopa at doses of at least 1200 mg/day suggests another form of parkinsonism. Dopamine ligand imaging can help distinguish between Parkinson disease (abnormal imaging findings) and drug-induced parkinsonism (normal imaging findings) or essential tremor (normal imaging findings). However, this imaging cannot discriminate between different types of degenerative parkinsonism such as Parkinson disease and dementia with Lewy bodies.

Causes of secondary or atypical parkinsonism can be identified by

- A thorough history, including occupational, medication, and family history
- Evaluation for neurologic deficits characteristic of disorders other than Parkinson disease
- Neuroimaging when patients have atypical features (eg, early falls, early cognitive impairment, ideomotor apraxia [inability to imitate hand gestures], hyperreflexia)

Treatment of Parkinson Disease

- Carbidopa/levodopa (mainstay of treatment)
- Amantadine, MAO type B (MAO-B) inhibitors, or, in few patients, anticholinergic medications
- Dopamine agonists
- Catechol *O*-methyltransferase (COMT) inhibitors, used in conjunction with levodopa, particularly when response to levodopa is wearing off

- Surgery if medications do not sufficiently control symptoms or have intolerable adverse effects
- Exercise and adaptive measures

A variety of oral medications are used to relieve symptoms of Parkinson disease ([1, 2]; see table Some Commonly Used Oral Antiparkinsonian Medications).

Levodopa is the most effective treatment. When Parkinson disease becomes severe, sometimes soon after diagnosis, response to levodopa can wear off due to disease progression, causing fluctuations in motor symptoms and dyskinesias. Early use of levodopa, however, does not hasten its ineffectiveness (3), and prompt initiation of levodopa often produces a noticeable improvement in quality of life.

Alternative oral medications include

- MAO-B inhibitors (selegiline, rasagiline)
- Dopamine agonists (pramipexole, ropinirole, rotigotine)
- Amantadine (best option when trying to decrease peak-dose [levodopa-induced] dyskinesias)
- Anticholinergics (trihexyphenidyl, benztropine)

Doses are often reduced in older adults. Medications that cause or worsen symptoms, particularly antipsychotics, are avoided.

Levodopa

Levodopa, the metabolic precursor of dopamine, crosses the blood-brain barrier into the basal ganglia, where it is decarboxylated to form dopamine. Coadministration of the peripheral decarboxylase inhibitor carbidopa prevents levodopa from being decarboxylated into dopamine outside the brain (peripherally), thus lowering the levodopa dosage required to produce therapeutic levels in the brain and minimizing adverse effects due to dopamine in the peripheral circulation.

Levodopa is most effective at relieving bradykinesia and rigidity and often substantially reduces tremor (2).

Common short-term adverse effects of levodopa are

- Nausea
- Vomiting
- Light-headedness

Common long-term adverse effects include

- Mental and psychiatric abnormalities (eg, delirium with confusion, psychoses, paranoia, visual hallucinations, punding [complex, repetitive, stereotyped behaviors])
- Motor dysfunction (eg, dyskinesias, motor fluctuations)

Hallucinations and paranoia occur most often in older adults and in patients who have cognitive impairment or dementia.

The dose that causes dyskinesias tends to decrease as the disease progresses. Over time, the dose that is needed for therapeutic benefit and the one that causes dyskinesia converge.

To initiate treatment, the dosage of carbidopa/levodopa is increased every 4 to 7 days as tolerated until maximum benefit is reached or adverse effects develop. The risk of adverse effects may be minimized by starting at a low dose and increasing the dosage slowly. Based on the patient's tolerance and response, clinicians can increase the dosage every week up to 2 or 3 tablets 4 or 5 times a day. Frequent dosing (4 to 5 doses a day) is recommended to decrease the effect of fluctuations in plasma levels of levodopa that can cause motor fluctuations and dyskinesias.

Most patients with Parkinson disease require 400 to 1200 mg a day of levodopa in divided doses every 2 to 5 hours. Preferably, levodopa should not be given with food because protein can reduce absorption of the medication.

If peripheral adverse effects of levodopa (eg, nausea, vomiting, postural light-headedness) predominate, increasing the amount of carbidopa may help. Carbidopa doses up to 150 mg are safe and do not decrease the efficacy of levodopa (3).

A variety of preparation forms of carbidopa/levodopa are available for specific uses, but none is superior to immediate-release carbidopa/levodopa 25/100 mg (1).

- A dissolvable immediate-release oral form of carbidopa/levodopa can be taken without water and is useful for patients who have difficulty swallowing. Doses are the same as for nondissolvable immediate-release carbidopa/levodopa.
- A controlled-release preparation of carbidopa/levodopa is usually used only to treat nighttime symptoms. When taken with food, absorption can be erratic, and it is present longer in the stomach than immediate-release forms.
- A formulation of levodopa/carbidopa intestinal gel can be given using a pump connected to a feeding tube inserted in the proximal small bowel. This formulation is used as treatment for patients who have severe motor fluctuations and/or dyskinesias that cannot be relieved by medications and who are not candidates for deep brain stimulation.

Occasionally, levodopa must be used to maintain motor function despite levodopa-induced hallucinations or delirium. In such cases, hallucinations and delirium can be treated with antipsychotic medications (oral quetiapine, pimavanserin, or clozapine). These medications, unlike other antipsychotics (eg, risperidone, olanzapine, all typical antipsychotics), do not aggravate parkinsonian symptoms. Although clozapine is most effective, its use is limited because agranulocytosis is a risk (estimated to occur in 1% of patients) and a complete blood count (CBC) must be closely monitored: weekly for 6 months, every 2 weeks for another 6 months, and every 4 weeks thereafter. However, the frequency may vary depending on the white blood cell (WBC) count.

Evidence suggests that the atypical antipsychotic pimavanserin is efficacious for psychotic symptoms and does not aggravate parkinsonian symptoms; also medication monitoring does not appear necessary (4, 5).

After 2 to 5 years of treatment, most patients experience fluctuations in their response to levodopa, and symptom control may fluctuate unpredictably between effective and ineffective (on-off fluctuations), as response to levodopa starts to wear off. Symptoms may occur before the next scheduled dose (off effects). The dyskinesias and off effects result from a combination of the pharmacokinetic properties of levodopa (particularly its short half-life as an oral medication), and disease progression.

Early in Parkinson disease, there are enough surviving neurons to buffer any oversaturation of dopaminergic receptors in the substantia nigra. As a result, dyskinesias are less likely to occur, and levodopa's therapeutic effect lasts longer because of the reuptake of excessive levodopa and its reutilization. As dopaminergic neurons are further depleted, each dose of levodopa saturates more and more dopamine receptors, resulting in dyskinesias and motor fluctuations because the delivery of levodopa to the substantia nigra becomes dependent on the plasma half life of levodopa (1.5 to 2 hours).

However, dyskinesias result mainly from disease progression and are not directly related to cumulative exposure to levodopa, as previously believed. Disease progression is associated with pulsatile administration of oral levodopa, which sensitizes and changes glutamatergic receptors, especially NMDA (*N*-methyl-D-aspartate) receptors. Eventually, the period of improvement after each dose shortens, and drug-induced dyskinesias result in swings from akinesia to dyskinesias. Traditionally, such swings are managed by keeping the levodopa dose as low as possible and using dosing intervals as short as every 1 to 2 hours, which are impractical. Alternative methods to decrease the off (akinetic) times include adjunctive use of dopamine agonists, as well as COMT and/or MAO inhibitors; amantadine may help manage dyskinesias.

Domperidone blocks peripheral dopamine receptors and does not cross the blood-brain barrier to affect the brain. By decreasing the decarboxylation of levodopa to dopamine, domperidone lessens the peripheral adverse effects of levodopa, thereby decreasing nausea, vomiting, and orthostatic hypotension. Domperidone, however, has been associated with serious cardiac risks and is not available in the United States except for investigational use.

Amantadine

Amantadine is most often used to:

- Ameliorate dyskinesias secondary to levodopa
- Lessen tremors

Amantadine is useful as monotherapy for early, mild parkinsonism but often loses its effectiveness after several months. Amantadine can also be used to augment levodopa’s effects by increasing dopaminergic activity, anticholinergic effects, or both. Amantadine can be helpful in managing dyskinesias secondary to levodopa.

Dopamine agonists

These medications directly activate dopamine receptors in the basal ganglia. They include

- Pramipexole

- Ropinirole
- Rotigotine
- Apomorphine

Oral dopamine agonists can be used as monotherapy but, as such, are rarely effective for more than a few years. Using these medications early in treatment, with small doses of levodopa, may be useful in patients at high risk of dyskinesias and on-off effects (eg, in patients < 60 years). Oral dopamine agonists may be useful at all stages of the disease, including as adjunctive therapy in later stages, but adverse effects may limit their use. Nearly 40% of patients using dopamine agonists develop impulse control disorders such as compulsive gambling, excessive shopping, hypersexuality, or overeating (6, 7). These adverse effects often require dose reduction or withdrawal of the causative medication and possibly avoidance of the medication class.

Pramipexole and **ropinirole**, given orally, can be used instead of or with levodopa in early Parkinson disease or can be added to treatment in advanced disease. These medications have a half-life of 6 to 12 hours and can be taken as immediate-release preparations 3 times a day. They can also be taken as sustained-release preparations once/day, helping minimize peaks and troughs in blood levels. Daytime sleepiness is a common adverse effect.

Rotigotine, given transdermally once a day, provides more continuous dopaminergic stimulation than medications given via other routes.

Apomorphine is a dopamine agonist used as rescue therapy when akinetic-rigid motor fluctuations are frequent and severe. It can be given sublingually or subcutaneously. Onset of action is rapid (5 to 10 minutes), but duration is short (60 to 90 minutes). Apomorphine can be given up to 5 times a day as needed. A 2-mg test dose is given first to check for orthostatic hypotension. Blood pressure (BP) is checked in the supine and standing positions before treatment and 20, 40, and 60 minutes afterward. Other adverse effects are similar to those of other dopamine agonists. Nausea can be prevented by starting trimethobenzamide 3 days before starting apomorphine and continuing trimethobenzamide for the first 2 months of treatment.

Apomorphine given by subcutaneous pump is available in some countries; it can be used instead of a levodopa pump in patients who have advanced Parkinson disease and who are not candidates for functional surgery.

Bromocriptine may be used in some countries, but in North America, its use is largely limited to treatment of pituitary adenomas because it increases the risk of cardiac valve fibrosis and pleural fibrosis.

Selective MAO-B inhibitors

Selective MAO-B inhibitors include selegiline and rasagiline.

Selegiline inhibits 1 of the 2 major enzymes that break down dopamine in the brain, thereby prolonging the action of each dose of levodopa. In some patients with mild off effects, selegiline helps prolong levodopa's effectiveness. Used initially as monotherapy, selegiline controls mild symptoms. Nonselective

MAO inhibitors yield amphetamine-like metabolites and, at higher doses, may cause a hypertensive crisis when foods containing tyramine (eg, some cheeses) are consumed;at the therapeutic dose listed, Selegiline remains a selective MAO-B inhibitor and dietary restriction is not required. Although virtually free of direct adverse effects, selegiline can potentiate levodopa-induced dyskinesias, mental and psychiatric adverse effects, and nausea, requiring reduction in the levodopa dose. Selegiline is also available in a formulation designed for buccal absorption (zydis-selegiline).

Rasagiline inhibits the same enzymes as selegiline. It is effective and well-tolerated in early and late disease; uses of rasagiline 1 mg orally once a day are similar to those of selegiline. Unlike selegiline, it does not have amphetamine-like metabolites, so theoretically, risk of a hypertensive crisis when patients consume tyramine is lower with rasagiline.

Anticholinergic medications

Anticholinergic medications can be used as monotherapy in early Parkinson disease and later to supplement levodopa. They are most effective for tremor. Doses are increased very slowly. Adverse effects may include cognitive impairment and dry mouth, which are particularly troublesome for older adults. Thus, anticholinergic medications are usually used only in young patients with tremor-predominant Parkinson disease or with some dystonic components. Rarely, they are used as adjunctive treatment in older adults who do not have cognitive impairment or psychiatric disorders.

Some studies using a mouse model indicate that use of anticholinergic medications should be limited because they appear to increase tau pathology and neurodegeneration; the degree of increase correlates with the medication's central anticholinergic activity (8, 9).

Commonly used anticholinergic medications include

- Benztropine
- Trihexyphenidyl

Antihistamines with anticholinergic effects (eg, diphenhydramine or pheniramine) are occasionally useful for treating tremor.

Anticholinergic tricyclic antidepressants (eg, amitriptyline at bedtime), if used for depression, may be useful as an adjunct to levodopa.

Catechol *O*-methyltransferase (COMT) inhibitors

COMT inhibitor medications (eg, entacapone, tolcapone) inhibit the breakdown of levodopa and dopamine and therefore may be useful as adjuncts to levodopa. They are used commonly in patients who have been taking levodopa for a long time when response to levodopa is progressively wearing off at the end of dosing intervals (known as wearing-off effects).

Entacapone can be used in combination with levodopa and carbidopa.

Tolcapone is a more potent COMT inhibitor because it can cross the blood-brain barrier; however, it is less commonly used because of the potential for liver toxicity. It is an appropriate option if entacapone does not sufficiently control off effects. Liver enzymes must be monitored periodically. Tolcapone should be stopped if alanine aminotransferase (ALT) or aspartate aminotransferase (AST) levels increase to twice the upper limit of the normal range or higher or if symptoms and signs suggest that the liver is damaged.

Opicapone is a third-generation COMT inhibitor that appears to be effective and safe in patients with Parkinson disease. Unlike tolcapone and like entacapone, opicapone does not require monitoring with periodic laboratory tests or multiple oral dosing.

TABLE

Some Commonly Used Oral Antiparkinsonian Medications

Medication	Major Adverse Effects
Dopamine precursors	
Carbidopa/levodopa 10/100, 25/100, or 25/250 mg (immediate-release or dissolvable)	**Central:** Drowsiness, confusion, orthostatic hypotension, psychotic disturbances, nightmares, dyskinesia **Peripheral:** Nausea, anorexia, flushing abdominal cramping, constipation, palpitations **With sudden discontinuation:** Neuroleptic malignant syndrome
Carbidopa/levodopa 25/100 or 50/200 mg (controlled-release; recommended only for nighttime [not daytime] symptoms)	
Antiviral medication	
Amantadine	Confusion, urinary retention, leg edema, elevated intraocular pressure, livedo reticularis **Rarely, with discontinuation or a decrease in dose:** Neuroleptic malignant syndrome
Dopamine agonists	
Apomorphine	Nausea, vomiting, light-headedness (due to orthostatic hypotension), hallucinations, impulse control disorder (gambling, buying, excessive eating, collecting) With the rotigotine patch only, redness, itching, and swelling in the legs and at the application site
Pramipexole	
Ropinirole	
Rotigotine (transdermal)	

Medication	Major Adverse Effects
Monoamine oxidase type B (MAO-B) inhibitors	
Rasagiline	Nausea, insomnia, somnolence, edema
Selegiline*	Possible potentiation of nausea, insomnia, confusion, and dyskinesias when given with levodopa
Anticholinergic medications†	
Benztropine	Drowsiness, dry mouth, urinary retention, constipation, blurred vision **Particularly in older adults:** Confusion, delirium, impaired thermoregulation due to decreased sweating
Trihexyphenidyl	
Catechol *O*-methyltransferase (COMT) inhibitors	
Entacapone‡	**Due to increased bioavailability of levodopa**: Dyskinesias, nausea, confusion, hallucinations **Unrelated to levodopa**: Back pain, diarrhea, changes in color of urine With tolcapone, risk of liver toxicity (rare)
Opicapone	
Tolcapone	

* Selegiline is also available in a formulation designed for buccal absorption.

† Anticholinergic medications should preferably not be used in older adults. Because these medications have adverse effects and because recent findings suggest that these medications may increase tau pathology and neurodegeneration, their use should be limited.

‡ Entacapone is also available in a triple combination tablet (carbidopa, levodopa, and entacapone).

Surgery

If pharmacotherapy is ineffective and/or has intolerable adverse effects, surgery, including deep brain stimulation and lesional surgery, may be considered.

Deep brain stimulation of the subthalamic nucleus or globus pallidus interna is often recommended for patients with levodopa-induced dyskinesias or significant motor fluctuations; this procedure can modulate overactivity in the basal ganglia and thus decrease parkinsonian symptoms in patients with Parkinson disease (10). For patients with tremor only, stimulation of the ventralis intermediate nucleus of the thalamus is sometimes recommended; however, because most patients also have other symptoms, stimulation of the subthalamic nucleus, which relieves tremor as well as other symptoms, is usually preferred. When the main problem is inadequate control of dyskinesias or when patients have an increased risk of cognitive decline, the globus pallidus interna is a good target.

Lesional surgery aims to stop overactivity directed to the thalamus from the globus pallidus interna; thalamotomy is sometimes done to control tremor in patients with tremor-predominant Parkinson disease. However, lesional surgery is not reversible and cannot be modulated over time; bilateral lesional surgery is not recommended because it can have severe adverse effects such as dysphagia and dysarthria. Lesional surgery involving the subthalamic nucleus is contraindicated because it causes severe ballismus.

Patient selection is the most important factor for successful functional surgery in Parkinson disease. Surgery is usually considered when pharmacotherapy of dyskinesias and/or motor fluctuations is ineffective or the effectiveness is severely limited. Pharmacotherapy may be inadequate because the medication has adverse effects that prevent further increases in the levodopa dose.

Other selection criteria include

- Parkinson disease present for 5 to 15 years
- Patient age < 70 years
- No significant cognitive decline, no affective disorder, and no terminal disease (eg, cancer, chronic renal failure, liver failure, significant cardiopathy), or uncontrolled diabetes mellitus or uncontrolled hypertension

Patients with cognitive impairment, dementia, or a psychiatric disorder are not suitable candidates for surgery because neurosurgery can exacerbate cognitive impairment and psychiatric disorders, and the risk of additional mental impairment outweighs the benefits of any improvement in motor function.

High-intensity focused ultrasound (HIFU)

MR-guided high-intensity focused ultrasound can be used to control severe tremor refractory to medications in patients with Parkinson disease. With this procedure, the ventral intermediate nucleus of the thalamus can be ablated with minimal risk of hemorrhage and infection, which may occur when more invasive neurosurgical procedures are used.

In addition to treatment of tremor, HIFU has shown promise in treating symptoms of bradykinesia and rigidity while reducing the rate of dyskinesias (11). Although HIFU is less invasive than other surgical interventions, deep brain stimulation is favored by most expert clinicians.

Physical and lifestyle measures

Maximizing activity is a goal. Patients should increase daily activities to the greatest extent possible. Physical or occupational therapy, which may involve a regular exercise program, may help promote physical conditioning. Therapists may teach patients adaptive strategies and suggest appropriate adaptations in the home (eg, installing grab bars to reduce the risk of falls).

To prevent or relieve constipation (which may result from Parkinson disease, antiparkinsonian medications, and/or inactivity), patients should consume a high-fiber diet, exercise when possible, and drink adequate amounts of fluids. Dietary supplements (eg, psyllium) and stimulant laxatives (eg, bisacodyl) can help.

Caregiver and end-of-life issues

Because Parkinson disease is progressive, patients eventually need help with normal daily activities. Caregivers should be directed to resources that can help them learn about the physical and psychological effects of Parkinson disease and about ways to help the patient function as well as possible. Because such care is tiring and stressful, caregivers should be encouraged to contact support groups for social and psychological support.

Eventually, most patients become severely disabled and immobile. They may be unable to eat, even with assistance. Because swallowing becomes increasingly difficult, death due to aspiration pneumonia is a risk. For some patients, a nursing home may be the best place for care.

Before people with Parkinson disease are incapacitated, they should establish advance directives, indicating what kind of medical care they want at the end of life.

Treatment references

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Key Points

- Parkinson disease is a synucleinopathy and thus can overlap with other synucleinopathies (eg, dementia with Lewy bodies, multiple system atrophy).
- Suspect Parkinson disease based on characteristic features: resting tremor, muscle rigidity, slow and decreased movement, and postural and gait instability.
- Distinguish Parkinson disease from disorders that cause similar symptoms based mainly on the history and physical examination results, but also test responsiveness to levodopa; sometimes neuroimaging is useful.
- Typically, use levodopa/carbidopa (the mainstay of treatment), but other medications (amantadine, dopamine agonists, MAO-B inhibitors, COMT inhibitors) may be used before and/or with levodopa/carbidopa.
- Consider surgical procedures, such as deep brain stimulation, if patients have symptoms refractory to optimal pharmacotherapy and do not have cognitive impairment or a psychiatric disorder.

Drug Information for the Topic